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Booklet Questions:-

Ex:-5:- [B.Q:-1]:- A sum of money becomes 1815/- @ 7%. SI after 3 yrs. Same sum becomes 2235/- @ same interest after 7 yrs. Find the sum?

Ans:- Sum = P = 100%

$A_{1-3} = 1815 = P + 3\text{ yrs SI} = 100\% + 3 \times 7\% = 121\%$

$\Rightarrow 100\% = \frac{1815}{121} \times 100 = \boxed{1500}$ (or)

$A_{1-7} = 100\% + 7 \times 7\% = 149\% = 2235$

$\Rightarrow 1\% = \frac{2235}{149} = 15 \Rightarrow \boxed{100\% = P = 1500}$

Ex:-6:- [B.Q:-2]:- Equal sums of money lent @ 4% p.a & 5% p.a for 2 yrs & 3 yrs. If one of them earns 245/- more interest than the other. Find the sum.

Ans:- Difference in Amounts in 2 cases = 245 (or)

Difference in interests in 2 cases = 245

P = 100% = same in both the cases.

$[3y \times 5\%] - [2y \times 4\%] = 245$

1st Interest - 2nd Interest

$\Rightarrow 15\% - 8\% = 7\% = 245 \Rightarrow 1\% = \frac{245}{7} = 35$

P = 100% = 3500/-

Ex:-7:- [B.Q:-3]:- A man invests 8000/- for 5 yrs @ 5% p.a. simple interests. Income tax 20% p.a on interest deducted at the end of each year. Find the amount at the end of 5th year?

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Ans:- P = 100%, R = 5%, I·T = 20% of 5% = 1% p.a.

Amount = 100% + 5×5% − 5×1% = 120%

= 120% of P = 120% of 8000 = 9600/-.

Ex:8:- [B.Q:-5]:- Taruun invested 10k on SI @ 8% p.a & another amount on SI @ 20% p.a. At the end of year the total interest on total amount is 12% P.a. The total investment?

Ans:- Assume total amount = x = total principle

1st Amount (princ) = 10k, 2nd Amount (princ) = x − 10k.

[10000 + 8% of 10k] + [x − 10000 + 20% of (x − 10000)] = x + 12% of x

1st A1

(or)

8% of 10k + 20% of (x − 10000) = 12% of x

800 + 0.2x − 2000 = 0.12x

0.2x − 1200 = 0.12x

0.2x − 0.12x = 1200 ⇒ 0.08x = 1200

⇒ $x = \frac{1200 \times 100}{8} = 15000$

Ex:-9:- [B.Q:-6]:- Vikram borrows a sum of 1500/- @ beginning of year. After 4 months 2100/- more is borrowed double the previous interest. At the end of year he paid total 416/- as interest. What is 1st rate of Interest?

Ans:- 1500/- borrowed for 1 year @ x% p.a (assume) & 2100 borrowed for $\frac{2}{3}$ yrs (1y − 4 months / 8 months) @ 2x% p.a.

x% of 1500 + $\frac{2}{3}$ × 2x % of 2100 = 416

$\frac{x}{100} \times 1500 + \frac{4x}{3 \times 100} \times 2100 = 416$

⇒ 15x + 28x = 43x = 416 ⇒ $x = \frac{416}{43} = \boxed{9.67}$ % p.a